Satya MARTHI

PhD Student in AI-Driven Control Systems

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About

PhD candidate in Control Systems and Reinforcement Learning specializing in reachability analysis, safe optimal control, and prescribed-time stability of nonlinear systems. Develops algorithms with Lyapunov-based guarantees and safety filtering. Strong implementation background in MATLAB and Python, integrating adaptive and disturbance-aware control for robotics and aerospace applications.

Education

2024–Present: **PhD, Control Systems Engineering**, *Universite de Lorraine, CRAN (Centre de recherche en automatique de Nancy)*, Nancy, France.

Funded by: Thesis funded by ANR (Agence Nationale de la Recherche) under the SOS (Self-Organizing, Smart) and safe heterogeneous robots fleet by collective emergence for a forest fire-fighting mission.

Work: Combining Control Systems and Reinforcement Learning for optimal control problems via safe reinforcement learning framework. Contributions to Hamilton–Jacobi reachability and prescribed-time stability analysis. Experienced in developing actor–critic and policy iteration algorithms for nonlinear autonomous systems.

2022–2024: **Master of Science, Aerospace Engineering**, *ISAE-SUPAERO (Institut supérieur de l'aéronautique et de l'espace)*, Toulouse, France.

Major: Aerospace systems and controls.

Course Work: System Identification and Estimation, Multi-variable Systems, Model Predictive Control, Optimal Control, Robust control, Non-Linear Control, Orbital Mechanics, Mission Analysis.

2017–2021: **Bachelor of Technology, Aerospace Engineering**, *Amrita Vishwa Vidyapeetham*, Coimbatore, India.

Course Work: Fundamentals of Aeroelasticity ,control theory, Aircraft structures, Aircraft Propulsions, Aerodynamics.

Skills

Technical Skills: MATLAB, Simulink, Python, PyTorch, ROS

Publications

Conference Papers

2025 ***Marthi S*, Jha MS, Kanso S, Ponsart JC, Theilliol D.**, *On-policy Safe Reinforcement Learning under Input Saturation and State Constraints for Nonlinear Discrete Time Systems*, In 2025 IEEE 64th **Conference on Decision and Control (CDC) 2025** (pp. 4817-4824), IEEE.

Experience

2024: **Research Internship**, *CRAN (Centre de recherche en automatique de Nancy)* , Nancy, France.
My objective as a research intern is to develop, evaluate, and use a control approach that combines the ideas of safe reinforcement learning and optimal control and adaptive control. The goal is to implement control-barrier functions while learning and applying the best control policy, resulting in online- optimal control and system safety

2022–2023 **Research Project**, *ISAE-SUPAERO (Institut supérieur de l'aéronautique et de l'espace)*, Toulouse, France.
Modelisation of Pair of tethered Aircraft for developing Control laws. The objective is to accurately model a system of pair of tethered aircraft and the construction of handling qualities and stability of a system of pair of tethered aircraft. Evaluating the impact of design parameters on stability and finally to determine a model of control laws to safely fly the aircraft.

2021–2022: **Internship**, *STAR (Space Technology and Aeronautical Rocketry)*, India.
My role as an Intern is to design a static test pad using Fusion-360, with good ergonomics and aesthetics, which can be assembled and disassembled easily. Static stress analysis was performed to ensure safety

Languages

French	Intermediate proficiency - B1
English	Fluent - C1
Telugu	Native language
Tamil	Fluent
Hindi	Fluent

Teaching Assistantship

2025-2026 **Deep Learning**, Polytech Nancy.

2025 **Automatic Control**, Polytech Nancy.

2025-2026 **Reinforcement Learning**, Polytech Nancy.

Referees

Dr. Didier Theilliol
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